BASE CHANGE BY AN ARBITRARY MORPHISM FOR A PERFECT PROPER MORPHISM OVER A NOETHERIAN BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0BZG (sources/stacks-duality/pinned-original-source.tex line 3332).

Allowed standard prerequisites (editorial).

- Support and sheafiness for a perfect proper morphism over Noetherian schemes. Same chapter, Lemma labelled lemma-proper-flat-noetherian-relative, cited at proof lines 3340-3342. Lets the functors be glued from open pieces.
- Checking the base change map over affine opens. Same chapter, Remark labelled remark-check-over-affines, cited at proof lines 3342-3344. Reduces to an affine morphism Y' -> Y.
- Base change in the affine proper Tor independent case. Same chapter, Lemma labelled lemma-more-base-change, cited at proof lines 3344-3346. Gives the isomorphism in the reduced case.

Source reference. Original equation 0AA6: sources/stacks-duality/pinned-original-source.tex, source lines 1192–1192. Same-chapter pointer into the author environment equation at source line 1192; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AAA: sources/stacks-duality/pinned-original-source.tex, source lines 3286–3286. Same-chapter pointer into the author environment lemma at source line 3286; invoked at proof line 3341 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0E9S: sources/stacks-duality/pinned-original-source.tex, source lines 1585–1585. Same-chapter pointer into the author environment remark at source line 1585; invoked at proof line 3344 of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA8: sources/stacks-duality/pinned-original-source.tex, source lines 1636–1636. Same-chapter pointer into the author environment lemma at source line 1636; invoked at proof line 3345 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to Y$ be a perfect proper morphism of Noetherian schemes. Let $g : Y' \to Y$ be a morphism with Y' Noetherian. If X and Y' are tor independent over Y, then the base change map (0AA6) is an isomorphism for all $K \in D_{QCoh}(\mathcal{O}_Y)$.*

Proof. By Lemma 0AAA formation of the functors a and a' commutes with restriction to opens of Y and Y'. Hence we may assume $Y' \to Y$ is a morphism of affine schemes, see Remark 0E9S. In this case the statement follows from Lemma 0AA8. □